Blessed G/Michael Catholic Secondary School

Semester 1 IT Mid1 Examination for Grade11

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade11 _____ **R.No:** _____

I. Match the items under Column A with the correct descriptions under Column B.

23. In SQL database queries, which command is used to extract records from a table matching specific criteria?

A. SELECT ... WHERE
B. UPDATE ... SET
C. INSERT INTO ... VALUES
D. DROP TABLE

II. Choose the best answer among the given alternatives.

____ **1.** Which layer of the OSI 7-layer reference model is responsible for routing data packets across disparate networks?

A. Data Link Layer **B.** Network Layer
C. Transport Layer **D.** Session Layer

____ **2.** Which transport layer protocol provides reliable, connection-oriented data transmission with error checking?

A. UDP **B.** TCP **C.** IP **D.** ICMP

____ **3.** In a Relational Database Management System (RDBMS), a primary key must be:

A. Unique and cannot contain NULL values
B. Numeric and allow duplicates
C. Text-based and optional
D. Shared between multiple tables without constraints

____ **4.** What is the primary role of the Domain Name System (DNS) in computer networking?

A. Encrypting outgoing passwords
B. Translating human-readable domain names into IP addresses
C. Compressing video streams
D. Filtering phishing emails

____ **5.** What is the domain of the real-valued rational function $f(x) = (2x + 1) / (x^2 - 16)$?

A. All real numbers except $x = 4$ and $x = -4$
B. All real numbers except $x = 16$
C. $x > 4$
D. All real numbers

____ **6.** If $f(x) = 2x - 3$ and $g(x) = x^2 + 1$, what is the composite value $(g \circ f)(2)$?

A. 2 **B.** 5 **C.** 10 **D.** 1

____ **7.** What is the inverse function $f^{-1}(x)$ of the one-to-one function $f(x) = 3x - 5$?

A. $f^{-1}(x) = (x + 5) / 3$ **B.** $f^{-1}(x) = (x - 5) / 3$
C. $f^{-1}(x) = 3x + 5$ **D.** $f^{-1}(x) = (x + 3) / 5$

____ **8.** Find the distance between the two points A(2, -3) and B(6, 0) in the Cartesian plane.

A. 5 units **B.** 7 units **C.** $\sqrt{7}$ units **D.** 25 units

____ **9.** What is the center and radius of the circle defined by the equation $(x - 3)^2 + (y + 2)^2 = 49$?

A. Center (3, -2), Radius = 7 **B.** Center (-3, 2), Radius = 7
C. Center (3, -2), Radius = 49 **D.** Center (-3, 2), Radius = 49

____ **10.** If $\sin(\theta) = 3/5$ and θ is in the first quadrant ($0 < \theta < \pi/2$), what is the value of $\cos(\theta)$?

A. 4/5 **B.** 3/4 **C.** 5/4 **D.** 1/5

____ **11.** What is the fundamental period of the trigonometric function $f(x) = \sin(4x)$?

A. $\pi / 2$ **B.** π **C.** 2π **D.** 4π

____ **12.** Which of the following lines is perpendicular to the line with equation y = 2x + 7?

A. y = -(1/2)x + 3 **B.** y = 2x - 5

C. y = (1/2)x + 1 **D.** y = -2x + 7

____ **13.** Solve the rational equation: (3 / x) + (2 / (x - 1)) = 0

A. x = 3/5 **B.** x = 5/3 **C.** x = -3/5 **D.** x = 1/2

____ **14.** Using trigonometric double-angle formulas, cos(2θ) is identically equal to:

A. $\cos^2(\theta) - \sin^2(\theta)$ **B.** 2 sin(θ) cos(θ)

C. $1 + 2\sin^2(\theta)$ **D.** $\cos^2(\theta) + \sin^2(\theta)$

____ **15.** What is the amplitude of the sinusoidal wave function y = -5 cos(3x - π)?

A. 5 **B.** -5 **C.** 3 **D.** π

____ **16.** The vertices of a triangle are located at (0, 0), (4, 0), and (0, 6). What is the area of this triangle?

A. 12 square units **B.** 24 square units

C. 10 square units **D.** 14 square units

____ **17.** Which of the following functions is an ODD function satisfying f(-x) = -f(x)?

A. $f(x) = x^3$ **B.** $f(x) = x^2 + 4$ **C.** f(x) = cos(x) **D.** f(x) = |x|

____ **18.** What are the coordinates of the midpoint of the segment joining (-4, 8) and (2, -2)?

A. (-1, 3) **B.** (-2, 6) **C.** (3, -1) **D.** (-1, 5)

____ **19.** Simplify the trigonometric expression: tan(x) * cos(x)

A. sin(x) **B.** sec(x) **C.** cot(x) **D.** 1

____ **20.** Which network device operates at Layer 2 (Data Link Layer) and forwards frames based on MAC addresses?

A. Network Switch **B.** Hub **C.** Router **D.** Repeater

____ **21.** Which protocol is used to automatically assign IP addresses and network configurations to host devices on a local network?

A. DHCP **B.** DNS **C.** FTP **D.** SMTP

____ **22.** Which twisted-pair ethernet cable standard uses an RJ-45 connector and supports gigabit data rates up to 1000 Mbps?

A. Category 5e / 6 (Cat5e/Cat6) **B.** Category 3 (Cat3)

C. Coaxial RG-59 **D.** Single-mode optical fiber

____ **24.** What is a Foreign Key in a relational database table?

A. A field in one table that uniquely identifies a row in another table (referential integrity)
B. An encryption key stored outside the database
C. A key that can only store foreign currency
D. A primary key that allows duplicate values

____ **25.** Which network security appliance monitors and controls incoming and outgoing network traffic based on predetermined security rules?

A. Firewall **B.** Modem

C. Network Interface Card (NIC) **D.** Patch Panel

____ **26.** Which of the following is an unguided (wireless) transmission medium?

A. Radio waves **B.** Fiber optic cable

C. Coaxial cable **D.** Shielded Twisted Pair (STP)

____ **27.** What is the default port number used by secure HTTPS encrypted web communication?

A. 443 **B.** 80 **C.** 21 **D.** 25

____ **28.** A Denial of Service (DoS) cyberattack primarily aims to:

A. Exhaust system resources and render a service unavailable to legitimate users
B. Steal hard drive passwords without notice
C. Physically destroy server motherboards
D. Format local user partitions

____ **29.** Which transmission mode allows data transmission in both directions simultaneously (e.g. telephone conversation)?

A. Full-Duplex **B.** Half-Duplex **C.** Simplex **D.** Multiplex

____ **30.** In database normalization, a table is in First Normal Form (1NF) if:

A. Every column contains atomic (indivisible) values and rows are unique
B. All non-key attributes depend on a foreign key
C. Transitive dependencies are eliminated
D. No indexes exist on the table

____ **31.** What does the acronym URL stand for?

A. Uniform Resource Locator **B.** Universal Routing Link

C. Unified Registry Line **D.** Unique Remote Locator

____ **32.** Which malware replicates itself across computer networks without requiring human intervention or host file attachment?

A. Worm **B.** Trojan Horse **C.** Macro Virus **D.** Spyware

____ **33.** Which network command is used to test reachability of a remote host and measure round-trip response time?

A. ping **B.** ipconfig **C.** tracert **D.** nslookup

____ **34.** The MAC (Media Access Control) address is hardcoded on which hardware component?

A. Network Interface Card (NIC) **B.** Hard disk drive controller

C. RAM module **D.** Power supply unit

____ **35.** Which cloud computing model delivers computing hardware, servers, storage, and networking over the internet on-demand?

A. IaaS (Infrastructure as a Service) **B.** SaaS (Software as a Service)

C. PaaS (Platform as a Service) **D.** DaaS